



Bias and Variance

Balancing underfitting and overfitting in machine learning

Simple explanations • worked examples • practical use cases



1. Bias

Bias is error caused by a model being too simple or relying on assumptions that cannot represent the real pattern.

High bias usually causes underfitting: the model performs poorly on both training data and new data.

Loan-default example

A model uses only income to predict default, ignoring debt, repayment history, utilization and employment stability. It may consistently miss important risk patterns.

- Symptoms: training and validation performance are both weak.
- Possible responses: add meaningful features, use a more capable model or reduce excessive regularization.

2. Variance

Variance is error caused by a model being too sensitive to the particular training dataset.

High variance usually causes overfitting: training performance is excellent, but performance on new data is poor.

A very deep decision tree may memorize unusual customer cases and noise rather than learning repeatable default patterns.

- Symptoms: a large gap between training and validation performance.
- Possible responses: simplify the model, collect more representative data, add regularization or use ensemble methods.

3. The trade-off

Expected prediction error $\approx$ Bias² + Variance + Irreducible noise

Condition	Training result	Validation result	Meaning
High bias	Poor	Poor	Underfitting
High variance	Very strong	Poor	Overfitting
Balanced	Strong	Strong	Good generalization

Increasing complexity can reduce bias but may increase variance. The goal is not maximum training accuracy; it is dependable performance on unseen cases.

4. Banking workflow and controls

Credit-risk modelling use case

- Split data into training, validation and test sets.
- Use cross-validation to estimate stability across different samples.
- Compare training and validation metrics, not training metrics alone.
- Tune model complexity and regularization using validation data.
- Evaluate once on the untouched test set.
- Monitor drift after deployment because customer behaviour can change.

Important distinction

Statistical bias of an estimator and unfair social bias are different concepts. A model can have low prediction bias yet still create unfair outcomes for a protected group. Both must be evaluated separately.

A useful model learns repeatable patterns, avoids memorizing noise and performs consistently on new customers.